

FREE PRECESSION OF NEUTRON STARS: ROLE OF POSSIBLE VORTEX PINNING

JACOB SHAHAM

Racah Institute of Physics, Hebrew University, Jerusalem, Israel

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ABSTRACT

Vortex pinning affects neutron stellar free precession by modifying its frequency and geometry, in a gyroscopic effect. For neutron stars with low oblateness, the free precession frequency is given by $\omega_p = \psi\Omega$, where ψ is the ratio of moments of inertia of the pinned superfluid to the rigid component of the rest of the star, and Ω is half the superfluid vorticity. This poses a severe difficulty on any interpretation of the 35^d cycle in Her X-1 in terms of neutron stellar wobble, if vortex pinning exists.

Subject headings: stars: neutron — stars: rotation — X-rays: binaries

I. INTRODUCTION

The possibility of vortex pinning to crustal nuclei has recently been considered with regard to superfluid neutron vortices in neutron stars (Anderson 1975; Anderson and Itoh 1975; Ruderman 1976; Anderson, Pines, and Shaham 1977). Preliminary estimates, based on neutron matter calculations of Hoffberg *et al.* (1970), Negele and Vautherin (1973), and Bäckman and Weise (1975), suggest that a maximal pinning force as large as 10^7 dynes can pin a vortex core to a lattice nucleus in certain regimes.

The angular rotation velocity of a superfluid, whose motion is governed by a vortex lattice, depends primarily on the geometry of that lattice. Pinning forces, by resisting any changes in vortex line positions, could force the vortex lattice to rotate at the crustal frequency, even when this frequency is different from that implied by the lattice geometry.

In this situation, most of the superfluid will move at an average angular velocity which differs from that of the vortex system. Considering this effect from the fluid point of view, one concludes that pinning provides the localized Magnus forces, which are always needed in order to have quantum vortex cores move at finite average velocity relative to the superfluid itself. From the point of view of a crustal pinning nucleus, fluid particles that pass by it suddenly acquire a vorticity; by exerting the required pinning force, the nucleus provides the necessary local acceleration, and also changes the velocity of the leaving fluid particles, to restore their irrotational motion.

The role of pinning forces in a decelerating star, in particular the strains they induce in the crust, have been discussed by Ruderman (1976) and by Anderson, Pines, and Shaham (1977), and some additional aspects are still under investigation. The purpose of the present paper is to point out that vortex pinning will play a very important role in neutron stellar free precession, regardless of the fine details of the microscopic calculations involved (which, to a large extent, have not yet been carried out). In realistic neutron stars, free precession will be modified by having both its frequency and its geometry change substantially. Thus, the microscopic phenomenon of pinning may have far-reaching macroscopic observational consequences in the neutron stars. In particular, if the neutron star in Her X-1 has its superfluid vortices pinned, it is unlikely that its free precessional frequency could be low enough to account for the 35^d on-off periodicity.

A preliminary report of this work has been published elsewhere (Shaham 1976*a, b*).

II. PINNING, CRACKING, AND THE MAGNUS FORCE

Consider the motion of a perfect fluid of density $\rho(\mathbf{r}, t)$ in a coordinate frame which rotates at frequency ω . Let $\mathbf{v}(\mathbf{r}, t)$ be the velocity field in that frame, in a situation where the vorticity $\boldsymbol{\zeta}' (\equiv \text{curl } \mathbf{v})$ is time-independent. Let $\mathbf{f}(\mathbf{r}, t)$ be the external force per unit volume exerted on the fluid. Taking the curl of the fluid Euler equation (see Landau and Lifshitz 1959), one finds

$$\text{curl} \left[\frac{1}{\rho} \mathbf{f} - \boldsymbol{\zeta} \times \mathbf{v} - \boldsymbol{\omega} \times \mathbf{r} \right] = 0, \quad (1)$$

here $\boldsymbol{\zeta} \equiv \boldsymbol{\zeta}' + 2\boldsymbol{\omega}$ is the fluid vorticity, as measured by an inertial observer.

We shall be interested only in cases for which $|\dot{\omega} \times r| \ll |\zeta \times v|$, so that the last term of (1) shall be ignored. Furthermore, when ζ is nonzero only in very localized regions, we may write the rotational part of f , the so-called Magnus force, as

$$f_M = \rho \zeta \times v. \quad (2)$$

For the quantum neutron vortices in neutron stellar crusts, f_M cannot be provided by gravity or pressure—these give only curl-free forces. Indeed, pinning forces are the only possibility.¹ One can estimate the maximum allowed value of v from the maximum pinning force available per nucleus, $F_p(r)$. Denote by 2Ω the average, $\langle \zeta \rangle$, of the superfluid vorticity, averaged over a macroscopic—but small—area, perpendicular to ζ . When integrated only over the cross section of a single vortex, we have

$$\int \zeta dS = \frac{h}{2m_n} n, \quad (3)$$

where m_n is the mass of the neutron and n is a unit vector in the direction of the vortex. Thus, we find for the total Magnus force needed per a vortex length of one internucleus spacing b ,

$$F_M \equiv \rho b \frac{h}{2m_n} \{n \times [(\Omega - \omega) \times r]\} \quad (4)$$

with the necessary condition

$$|F_M| \leq F_p. \quad (5)$$

At the typical values of $\rho \sim 3 \times 10^{13} \text{ g cm}^{-3}$, $b \sim 5 \times 10^{-12} \text{ cm}$, $r \sim 10^6 \text{ cm}$, $F_p \sim 10^7 \text{ dynes}$, (4) and (5) yield

$$|\Omega - \omega| \leq 50 \text{ s}^{-1}. \quad (6)$$

The fastest neutron star known hitherto, the Crab pulsar, has a rotation angular frequency of $\Omega \approx 200 \text{ s}^{-1}$; even for this fast neutron star, (6) shows vortices to remain pinned even when free precession tilts ω away from Ω ($\sim \omega$) by $\sim 15^\circ$.

There exists another process by which vortices can move relative to the bulk of the crust, arising from the fact that the maximal pinning forces $F_p(r)$ vary with radius in the star. This variation, as pointed out by Ruderman (1976), will give rise to shear strains, which could cause crustal cracking. During such cracking, vortices could move to their hydrodynamical positions, to be repinned until the next cracking.

The geometry and dynamics of this cracking processes have still not been worked out in any great detail; we hope to return to discuss this problem in a later publication. For the purpose of this paper, we shall ignore cracking altogether, and confine ourselves to free-precessional motion occurring *between successive quakes*. The actual, long-term, behavior of free precession may be interrupted by quakes; but as long as vortices are tied to the crust for any finite period, all of the conclusions of this paper will hold during these periods.

III. MODEL CALCULATIONS

We consider a rotating neutron star, in which part of the superfluid neutron vortices are constrained to move rigidly with the stellar crust. This constraint can come about either by local, direct, pinning of the vortex or by the indirect effect of distant pinning, transmitted along the whole vortex via the “rigidity” of the vortex lattice (Ruderman and Sutherland 1974). In the first case, the vortices in the crust are decoupled from the core vortices, while in the second they are not.

Let us consider the case in which the angular momentum L_f of the constrained superfluid also moves rigidly with the crust; we shall see below that an almost spherical symmetry of the constrained superfluid guarantees that to be a good approximation.

Let us divide the neutron star into two parts: the constrained superfluid and the rest of the star (henceforth called “the star”), which we shall assume first to be infinitely rigid. Let the “star” have an axially symmetric moment of inertia tensor, with principal values A , A , and C ($> A$). Let $\mathcal{F}$ be a coordinate frame whose principal directions, i , j , and k , coincide with the principal directions of the “stellar” moment of inertia tensor. If ω is the “stellar” angular frequency, then the angular momentum of the “star,” L_c , is given in $\mathcal{F}$ by

$$L_c = \begin{vmatrix} A & & \\ & A & \\ & & C \end{vmatrix} \omega. \quad (7)$$

¹ Note, that pinning forces act only on the *normal* part of the neutron fluid. Therefore, even though the local pinning force, which acts between a given nucleus and a given vortex core, can be written as the gradient of the superfluid gap energy, its overall effect on the fluid is *not* curl free, as it depends both on r and on $\zeta(r)$. Further note that even with pinning at particular points on vortex lines there can be some vortex line bending in response to Magnus forces. This can play a role analogous to that of crustal rigidity.

We now write the Euler equations in $\mathcal{F}$ for the total angular momentum,

$$L_c \dot{\dagger} + L_f \dot{\dagger} + \boldsymbol{\omega} \times L_c + \boldsymbol{\omega} \times L_f = 0, \quad (8)$$

where the dagger denotes time differentiation in $\mathcal{F}$, and impose on them the constraint $L_f \dot{\dagger} \equiv 0$. The resulting equation,

$$L_c \dot{\dagger} + \boldsymbol{\omega} \times L_c = -\boldsymbol{\omega} \times L_f, \quad (9)$$

is an Euler equation for L_c , with an extra effective driving torque $N_{\text{eff}} = -\boldsymbol{\omega} \times L_f$. In the spherical configuration, this is simply the total torque acting on the “star” due to the Magnus force.

To get some insight into the motion of L_c , we first consider the case of $L_f \parallel k$, where k is the “stellar” symmetry direction (see Fig. 1). In this case (which is not an unlikely situation in realistic neutron stars), L_f is rotating in space around the (fixed) direction of the total angular momentum, $L \equiv L_c + L_f$. To see this, note that L_f is assumed to rotate rigidly with the “star,” so that in an inertial frame F we have

$$\frac{d}{dt} L_f = \boldsymbol{\omega} \times L_f = (\omega_3 k + \boldsymbol{\omega}_-) \times L_f = \boldsymbol{\omega}_- \times L_f = \frac{1}{A} L_{c-} \times L_f = \frac{1}{A} L_c \times L_f = \frac{1}{A} L \times L_f, \quad (10)$$

where $\boldsymbol{\omega}_-$ and L_{c-} denote components in the (i, j) -plane, and d/dt denotes time differentiation in F .

To see how the rest of the “star” rotates, we jump on the frame $\mathcal{C}$ which rotates around L at frequency L/A ; in $\mathcal{C}$, L_f is a constant vector too, and therefore $\mathcal{C}$ differs from $\mathcal{F}$ only by a rotation around L_f (i.e., around k). This

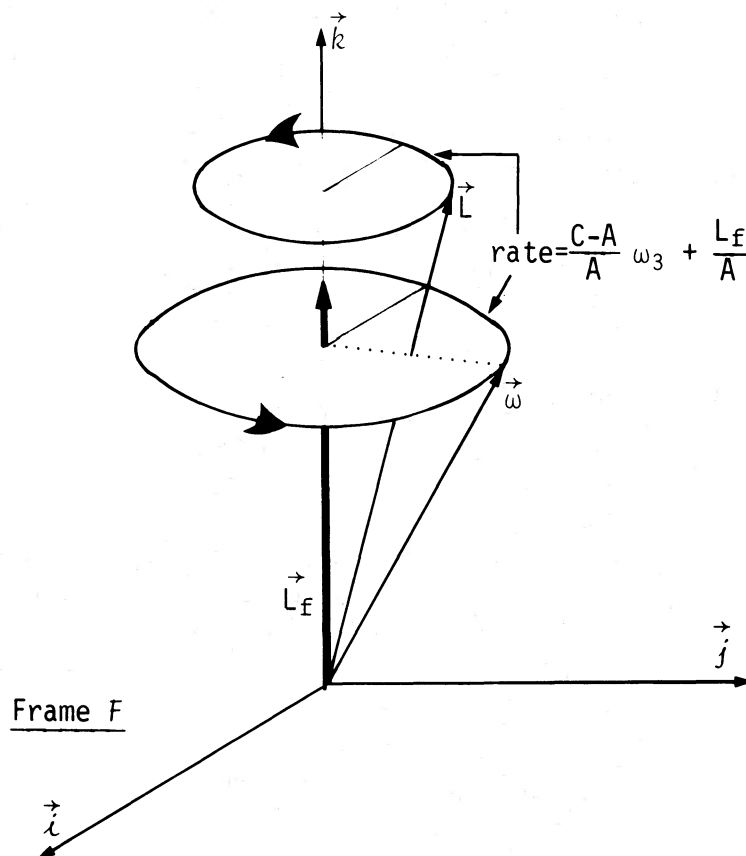


FIG. 1a

FIG. 1.—Free precession for the case $L_f \parallel k$: (a) In the corotating frame $\mathcal{F}$; $\boldsymbol{\omega}$, L_c , and L rotate around k (and L_f) at frequency $[(C - A)/A]\omega_3 + L_f/A$. (b) In an inertial frame F ; k (and L_f) rotate rigidly around L at frequency L/A , while the whole star rotates also around k at frequency $-[(C - A)/A]\omega_3 + L_f/A$.

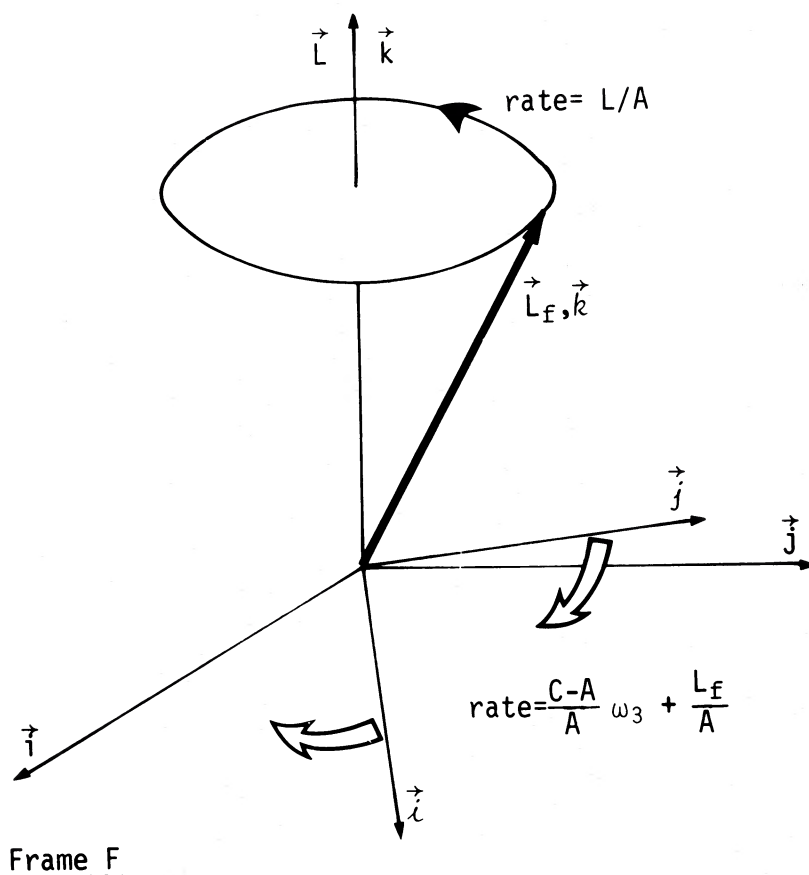


FIG. 1b

FIG. 1.—Continued

rotation is the classical “free precession” or “wobble,” and we calculate its frequency vector, ω_p , directly as follows:

$$\omega_p \equiv \omega - \frac{L}{A} = \omega_3 k + \omega_- - \frac{L_{C3}k}{A} - \frac{L_{C-}}{A} - \frac{L_f}{A} = -\left(\frac{C-A}{A}\omega_3 + \frac{L_f}{A}\right)k. \quad (11)$$

Were there no constrained superfluid ($L_f \equiv 0$), (11) would give the ordinary free precession frequency for rigid axisymmetric tops. With $L_f \neq 0$, ω_p is seen from (11) to be modified.

We also note that an $\mathcal{F}$ observer sees the motion discussed above as a rotation of L (as well as L_c and ω) around k , at frequency $-\omega_p$.

We see this directly from (9), for

$$\begin{aligned} -L\dot{\dagger} &= \omega \times L = \omega \times L_c + \omega \times L_f = \omega_3 k \times L_{c-} + \omega_- \times L_{c3}k + \omega_- \times L_f \\ &= \omega_- \times (-A\omega_3 + C\omega_3 + L_f)k = L \times \left(\frac{C-A}{A}\omega_3 + \frac{L_f}{A}\right)k = \omega_p \times L. \end{aligned} \quad (12)$$

One can thus conclude that the case $L_f \parallel k$ differs from ordinary free precession of an axisymmetric top only by the different frequency. The geometry of the motion stays the same, as axial symmetry was not broken.

In the case $L_f \not\parallel k$ (or in the equally complicated case of a triaxial star, which we shall not discuss here specifically), the motion of ω (and the “star”) becomes rather complicated for the external observer on F . He is no longer able to describe it as a simple superposition of two rotations. However, for an observer on $\mathcal{F}$, the motion of ω (and the other relevant vectors) still remains periodic, and we shall therefore analyze the motion in $\mathcal{F}$.

From (9), we find in $\mathcal{F}$ two constants of motion: One is the magnitude of L ,

$$L^2 = (L_{f1} + A\omega_1)^2 + A^2\omega_2^2 + (L_{f3} + C\omega_3)^2. \quad (13)$$

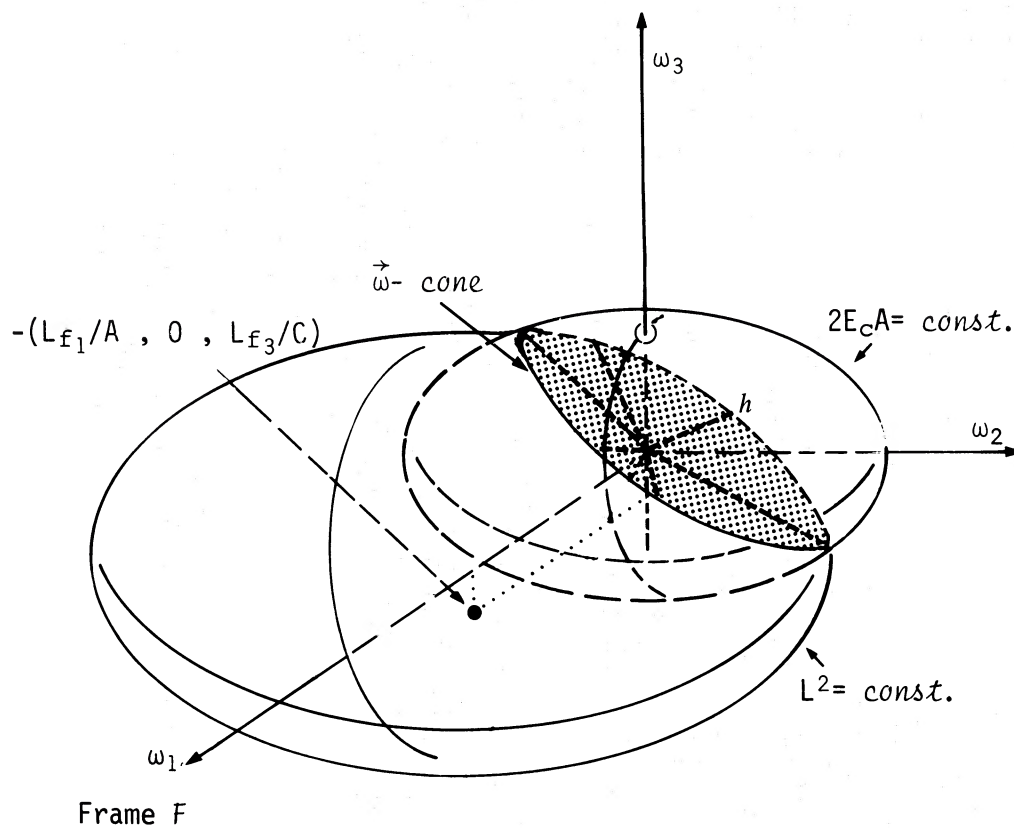


FIG. 2.—The surfaces of constant L^2 and $2E_c A$ in ω -space, with coordinate axes $\mathcal{F}$, for $L_f \parallel k$. An intersection curve h represents allowed ω values, and ω sweeps h while free precessing. For convenience of drawing, $A < C$ in this figure.

(Without loss of generality, we have chosen $\mathcal{F}$ so as to have $L_{f2} \equiv 0$.)

The other constant is the “stellar” kinetic energy E_c ,

$$2E_c = A(\omega_1^2 + \omega_2^2) + C\omega_3^2. \quad (14)$$

Figure 2 shows the body-fixed ω -space, with the surfaces of constant L^2 and constant E_c drawn in it. Both are ellipsoids of revolution around the ω_3 axis; the E_c ellipsoid is centered at the origin, while the L^2 ellipsoid is centered around $(-L_{f1}/A, 0, -L_{f3}/C)$. The curves of intersection of the two surfaces represent the allowed ω values in $\mathcal{F}$. If and when such values exist, they group in one or two closed curves, depending on L_f and on the E_c and L^2 values.

Let h be one such curve, on which ω happens to lie at a certain instant. Then ω will span h , and only h , in a periodic manner. To show that, we note that the Euler differential equation (9) is of first order in ω . Therefore, the velocity of the representative point of ω on h is determined only by its position; thus, the representative point will not stop on h to turn back and to cross identical points in a different sense; hence, the motion along h must be periodic.

On connecting all the points on h with the origin we obtain the “ ω cone”; ω is moving on the surface of that cone.

One is able to “orient” oneself in this picture by considering the simple cases:

For $L_f \parallel k$, the two ellipsoids of revolution are centered on the k axis, and the ω cone is circular, with k as a symmetry axis.

For given E_c and L^2 , there are L_f values for which there is only one intersection curve, so that ω_3 can only have one value. Different L_f values can produce two curves (or none) depending on $E_c A/L^2$.

For $A = C$, h is the (single) intersection curve of two spheres. The circular ω cone has the L_f direction as a symmetry direction. For given E_c and L^2 , no motion is possible for too large an L_f . For smaller ones, $\omega \cdot L_f$ is positive or negative depending on $E_c A/L^2$.

The exact temporal behavior of ω can be found on incorporating E_c and L^2 into (9). Some simple algebra gives ($L_{f1} \neq 0$),

$$t - t_0 = \left(\frac{2A}{E_c}\right)^{1/2} \int \frac{dx}{\sqrt{[\mathcal{P}(x)]}}, \quad (15)$$

where the integration is from $C\omega_3(t_0)/\sqrt{(2E_cA)}$ to $C\omega_3(t)/\sqrt{(2E_cA)}$;

$$\omega_1 = \frac{2E_cAD - 2C\omega_3L_{f3} - \epsilon C^2\omega_3^2}{2L_{f1}A}, \quad (16)$$

$$\omega_2 = \left(\frac{2E_c}{A} - \frac{C}{A}\omega_3^2 - \omega_1^2\right)^{1/2}, \quad (17)$$

where in (15)–(17) we have defined

$$D = \frac{L^2 - L_f^2 - 2E_cA}{2E_cA}, \quad (18)$$

$$\epsilon = (C - A)/C, \quad (19)$$

and

$$\mathcal{P}(x) = -\left\{D - x\left[\epsilon x + L_{f3}\left(\frac{2}{E_cA}\right)^{1/2}\right]\right\}^2 + \frac{2L_{f1}^2}{E_cA}[1 - (1 - \epsilon)x^2]. \quad (20)$$

The period of the stellar free precession will be given by

$$\frac{2\pi}{\omega_p} = \left(\frac{8A}{E_c}\right)^{1/2} \int_{\alpha}^{\beta} \frac{dx}{\sqrt{[\mathcal{P}(x)]}}, \quad (21)$$

where α and β are the two zeros of $\mathcal{P}(x)$ between which $\mathcal{P}(x) > 0$.

Equation (21) can, in principle, be evaluated by means of elliptic integrals. The main practical obstacle is the calculation of α and β , though numerical methods are straightforward. One may, again, acquire some feeling for the ω_p values by considering two extremities:

For $L_f \parallel k$, (21) indeed reduces to the ω_p value of (11) (note that when $L_{f1} \neq 0$, $|\omega_p|$ does not have a meaningful relation with ω_p).

For $L_f \perp k$, we find

$$\frac{2\pi}{\omega_p} = \frac{8A}{\sqrt[4]{(4b)}} F\left[\frac{\pi}{2}, \frac{(b^{1/2} - a)^{1/2} 2^{1/2}}{\sqrt[4]{b}}\right], \quad (22)$$

where

$$a = 2L_f^2 - \epsilon(L^2 + L_f^2 - 2E_cA), \quad (23)$$

$$b = 4L_f^4 - 4(L^2 + L_f^2 - 2E_cA)\epsilon L_f^2 + 4L_f^2L^2\epsilon^2 \quad (24)$$

and the elliptical integral F is defined, in the standard fashion, by

$$F(\tfrac{1}{2}\pi, k\uparrow) = \frac{\pi}{2} \left[1 + \frac{1^2}{2^2} k\uparrow^2 + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} k\uparrow^4 + \dots\right]. \quad (25)$$

For $\epsilon(L^2 + L_f^2 - 2E_cA)/L_f^2 \ll 1$, (22) gives $\omega_p \sim L_f/A$. This is also the approximate value for ω_p when $L_f \parallel k$ and for $\epsilon \ll 1$. Thus, one concludes that if ϵ is only small enough, then

$$\omega_p \sim L_f/A \equiv \psi\Omega \quad (26)$$

for any L_f direction. In (26), ψ is the ratio between the constrained superfluid and the stellar moments of inertia, and Ω is the rotation frequency of the constrained superfluid (half its vorticity).

IV. ROLE OF FINITE CRUSTAL RIGIDITY

Free precession is a consequence of a misalignment between the angular velocity vector and the angular momentum. Finite rigidity of precessing bodies reduces the infinite-rigidity free precessional frequencies (Pines and Shaham 1972), as the angular momentum and angular velocity vectors tend to be more aligned, due to the solid distortions.

When considering the role of finite rigidity in presence of pinning, we immediately notice, that if the misalignment due to the constancy of L_f is much larger than misalignment due to stellar oblateness, then finite rigidity will be of little importance; in the complementary (not much larger) case, however, finite rigidity will modify free precession both by the very distortion of the star and by the influence of the latter on modifying L_f .

Detailed calculations for the latter case require, again, knowledge of boundary behavior, and are thus not yet possible. However, for realistic neutron stars, the first case is the important one: The misalignment due to L_f is dominant unless a solid core exists; and in the latter case, the neutron star is very rigid anyway. The model calculation is therefore quite suitable for realistic neutron stars.

V. WOBBLE DAMPING

Wobble damping in neutron stars is mostly due to fluid-solid coupling. The canonical model (Baym *et al.* 1969) has the free vortex-cores exchange linear momentum, in a direction perpendicular to their length, with the rigid rotator (consisting of the solid and charged particles) on a time scale τ , which is the "postglitch" relaxation time.

It can be shown (Shaham 1975), that essentially similar relaxation processes contribute as well to the damping of free precession, even though in this case momentum transfer along vibrating vortices plays some role too. In a steadily rotating neutron star, the resulting damping time τ_w is of the order of $\tau\omega/\omega_p$ (Shaham 1975; Baym, Lamb, and Lamb 1975), while the existence of accelerating axial torques modify this time to be the smallest of $\tau\omega/\omega_p$ and the acceleration time scale.

Vortex pinning modifies wobble damping by dramatically reducing—essentially to zero—the relaxation time for the pinned vortices. In the case for which crustal direct pinning induces effective pinning throughout the length of the vortex, pinning would involve the complete disappearance of the fluid-solid long relaxation time. When core vortices are decoupled and move independently, the only effect of pinning in realistic neutron stars, for which $\omega\tau \gg 1$, comes from the decrease in τ_w due to the large rise in ω_p . It should be noted that the Q -value for the wobble model depends only on $\omega\tau$, and remains essentially unchanged.

However, wobble damping via crust-core coupling, in the situation where the core superfluid is not pinned to the crust, will also have the effect of modifying (11): To confirm that, let us retain the notation of equation (7) for describing the rigid component and denote by L_n the angular momentum of the core superfluid. L_f will still refer to the pinned superfluid, which now specifically denotes crustal superfluid only. For simplicity, let the core superfluid have a spherical moment-of-inertia tensor of principal value I , and let the dissipative crust-core relaxation be independent of the direction of their relative angular frequency.

Under these circumstances, we replace equations (8) and (9), in the crustal corotating frame, by

$$L_c\dot{\dagger} + L_f\dot{\dagger} + L_n\dot{\dagger} + \omega \times (L_c + L_f + L_n) = 0, \quad L_f\dot{\dagger} = 0, \quad L_n\dot{\dagger} = -\alpha(L_n/I - \omega) - \omega \times L_n. \quad (27)$$

Define

$$1/\tau = \alpha/I + \alpha/C, \quad (28)$$

$$f = I/(I + C), \quad (29)$$

$$\eta = (C - A)/A, \quad (30)$$

$$L_{n1} + iL_{n2} \equiv I \frac{1 + \chi}{1 + \eta} (\omega_1 + i\omega_2) \text{ (defining } \chi), \quad (31)$$

$$\hat{\Gamma} = \omega_1 + i\omega_2. \quad (32)$$

We write equations (27), with definitions (28)–(32), in form of the following three integro-differential equations, which are valid for times $t \gg \tau$:

$$\begin{aligned} \chi\dot{\dagger} + i\omega_3(1 + \eta)\chi - \frac{1}{\tau}(\eta - \chi)(1 + f\chi) + i(1 + \chi)\left(\frac{L_{f3}}{A} - \frac{\omega_3 L_{f1}}{A\hat{\Gamma}}\right) \\ = \int_0^t \exp\left(-\frac{t-\theta}{\tau}\right) \left(\frac{\hat{\Gamma} - \hat{\Gamma}^*}{2A} L_{f1} - i|\hat{\Gamma}|^2 \text{Im } \chi\right) d\theta, \end{aligned} \quad (33)$$

$$\begin{aligned} (1 + \eta)\omega_3 = (1 + \eta)[(1 - f)\omega_3 + f\omega_{n3}]_{t=0} - f \int_0^t \left[1 - \exp\left(-\frac{t-\theta}{\tau}\right)\right] |\hat{\Gamma}|^2 \text{Im } \chi d\theta \\ + i \frac{L_{f1}}{2A} \int_0^t (\hat{\Gamma} - \hat{\Gamma}^*) \left[1 - f + f \exp\left(-\frac{t-\theta}{\tau}\right)\right] d\theta, \end{aligned} \quad (34)$$

$$\frac{d\hat{\Gamma}}{dt} = i \left[\left(\eta\omega_3 + \frac{L_{f3}}{A} \right) + i \frac{f}{\tau} (\eta - \chi) \right] \hat{\Gamma} - i\omega_3 \frac{L_{f1}}{A}. \quad (35)$$

We are essentially interested in finding χ , whose real part will determine the damping time and whose imaginary part will determine the wobble frequency, by (35).

To do this we first note that in the limit of zero coupling ($\alpha \rightarrow 0$, $\tau \rightarrow \infty$), (27) admits two independent modes of motion: In the first mode, the crust is freely precessing with the frequency (11) (note, however, that here A denotes only crustal-nuclei moment of inertia, not the “stellar” one) while $\mathbf{L}_n$ is always in the direction of $\boldsymbol{\omega}$; in the second mode, $\boldsymbol{\omega} \parallel \mathbf{k}$ while $\mathbf{L}_n$ stays constant in space, and is thus seen to rotate at the constant frequency $-\boldsymbol{\omega}$ in the corotating frame.

On “turning the coupling on,” these two modes are modified: In the first, $\mathbf{L}_n$ lags slightly behind $\boldsymbol{\omega}$ ($\chi \neq \eta$ and becomes imaginary); in the second mode, $\boldsymbol{\omega}$ does not stay in the direction of $\mathbf{k}$ but is dragged away from it by $\mathbf{L}_n$ via their coupling. The second mode can be shown to be damped on the “postglitch” time scale. Therefore, solving (33)–(35) for times $t \gg \tau$ will reproduce the free precession mode, which will survive after such times provided $\tau_w \gg \tau$ and $t \lesssim \tau_w$. Note, that if the condition $\tau_w \gg \tau$ is not satisfied, one is unable to speak of a single-frequency precession mode; from (35) it is clear, that under these circumstances, ω_p will be time-dependent over the whole lifetime of the free precession, and will lose its identity.

The equations are not readily solvable in all cases. We shall concentrate here, however, on their solution under the conditions²

$$\omega_p \ll \omega \quad (36)$$

and

$$\omega\tau \gg 1. \quad (37)$$

These are the relevant conditions in realistic neutron stars, and are also the simplifying conditions for our mathematical problem.

Let us first assume, again, that $L_{f1} = 0$, $L_{f3} = L_f$. As will become self-consistently clear below, conditions (36)–(37) guarantee that $|\chi| \ll 1$ and that both χ and $|\Gamma|$ vary slowly over times of order τ . Hence, we can replace the operator

$$\int_0^t d\theta \exp[(t - \theta)/\tau]$$

by τ in (33) and (34), and neglect $\chi^\dagger$ with respect to χ/τ in (33). To lowest order we then find,

$$\text{Re } \chi \approx \frac{|\Gamma|^2}{\omega^2} \eta - \psi \frac{\omega_3 \Omega}{\omega^2} \quad [o(1)], \quad (38)$$

$$\text{Im } \chi \approx -\frac{1}{\omega\tau} \left(\frac{\omega_p}{\omega} \right) \quad [o(1)], \quad (39)$$

$$\omega_p \approx \eta\omega_3 + \psi\Omega, \quad (40)$$

$$\tau_w \equiv -\frac{|\Gamma|}{d|\Gamma|/dt} \approx \frac{\tau}{f} \left(\frac{\omega}{\omega_p} \right) \left(\frac{\omega}{\omega_3} \right). \quad (41)$$

Clearly, $\tau_w \gg \tau$, showing that the free precession mode indeed survives on times $t \gg \tau$, for which our solution is valid. Both χ and $|\Gamma|$ vary on this time scale, τ_w .

Let us now assume $\eta = 0$, and put $\mathbf{L}_f$ in the $\mathbf{k}$ direction. A brief examination of (33)–(35) under these conditions reveals that the solution is again simply given by (38)–(41), with $\eta = 0$.

The general case has a rather complicated behavior; however, under our previous assumptions (leading to [26]) regarding the smallness of η , no qualitatively different behavior is expected, and $\hat{\Gamma}$ will precess at ω_p with a motion similar to that described in Figure 2, until, on a time scale τ_w , it will relax to its steady-state position,

$$\hat{\Gamma} = \frac{\omega_3(L_{f1}/A)}{\eta\omega_3 + L_{f3}/A} \quad (42)$$

for the appropriate ω_3 value. Equation (42) is, indeed, the value one finds when solving (13)–(14) under the requirement that the curve h degenerates into a single point. The star, while conserving the total angular momentum, will dissipate just enough kinetic energy to arrive at (42) after a time scale $\sim \tau_w$.

² Condition (36) implies $\eta, \psi \ll 1$, unless one encounters the very unique situation in which $\omega_p \ll \min(\eta\omega, \psi\Omega)$. We shall not consider this case here.

VI. ROLE OF NONSPHERICAL CAVITY

The fluid angular momentum L_f depends, under steady-state conditions, both on the fluid (constant) vorticity, 2Ω (directed along the vortex lines), and the stellar rotation vector, ω . For a perfect fluid rotating inside of a rigid cavity, the latter dependence appears via any departure from sphericity of the inner cavity boundary (Moiseyev and Romyantsev 1968). Its physical origin is traced to the necessity to avoid any nonzero relative velocity of the fluid component normal to that boundary. This is achieved, in steady state, by the appearance of a potential flow, which modifies the angular momentum whenever $\Omega - \omega \neq 0$. When the boundary is elliptical, the modification is of second order in its oblateness.

For realistic neutron stars, the rigid cavity model is very likely not a very good one: First, the external surface, which bounds the regime of neutron superfluidity, as well as any possible inner such surface, will not remain constant while ω is changing due to changes in density and, to a lesser extent, temperature. Furthermore, even when these surfaces do not change very much, neutron fluid can certainly flow across them, hence producing a rather different potential flow.

As in the case of the role of finite rigidity, detailed calculations of the potential flow require yet unavailable details of the structure of the boundary. First approximation would suggest that the potential flow component will diminish, due to the penetrability of the boundary; hence in realistic neutron stars our model calculation will remain a good approximation also with regard to this effect.

VII. THE NEUTRON STAR IN HERCULES X-1

An immediate consequence of the preceding discussion relates to the suggestion (Brecher 1972; Pines, Pethick, and Lamb 1973), that the 35^d on-off periodicity of the Her X-1 system (Giacconi *et al.* 1974) is due to free precession of the underlying neutron star. Lamb *et al.* (1975) have shown that, if the 35^d period was indeed free precession, then there was a possibility of pumping it via the modulation it induced on the accretion flow, so that a sufficiently large amplitude was a definite astrophysical possibility.

However, if pinning of vortices takes place in that neutron stellar crust, it is impossible to have a free precessional frequency as low as $2\pi/35 \text{ day}^{-1}$, with the measured rotation period of 1.24 s: either free precession is constantly interrupted by pinning-induced starquakes and retains no phase coherence or, if quaking is less frequent, the observed period is limited by equation (26) since the expected ϵ value for Her X-1 is sufficiently small (Pines, Pethick, and Lamb 1973). Let us discuss this limit.

There is now some evidence that the mass of the neutron star in Her X-1 is $(1.33 \pm 0.13) M_\odot$ (Middleditch, Mast, and Nelson 1974). One might want to estimate ψ for such a star from recent neutron stellar models.³ First, we assume the crust to rotate rigidly with the core, i.e., infinite core-crust coupling and no pinning in the core. Following a review on neutron star structure by Arnett and Bowers (1974), Ruderman (1976) finds, depending on the model,

$$\psi \approx (1-3) \times 10^{-2}. \quad (43)$$

We can also estimate ψ for such a neutron star using the tensor-interaction model, following Pandharipande, Pines, and Smith (1975), to be even closer to unity. On the other hand, if another situation exists, in which the superfluid is pinned through the core, ψ will again be closer to unity than estimated from equation (43). Finally, we note that the opposite case of zero core-crust coupling, when the core superfluid is fully decoupled from the rest of the star, ψ involves, in the denominator, only the crustal moment of inertia, and thus becomes even larger.

It seems, therefore, safe enough, to write

$$\psi \gtrsim 10^{-2} \quad (44)$$

for any existing model of stellar structure or of pinning.

If $\Omega \sim \omega$, and ω is the present value of the Her X-1 rotation, (26) and (44) combine to give an upper bound for the wobble period P_w , which is

$$P_w \lesssim 120 \text{ s}. \quad (45)$$

Clearly, the value of Ω need not be equal to the current value of ω ; it will, however, be related to it via equation (6). Thus, we could have $\Omega \sim 55 \text{ s}^{-1}$, which will increase the upper limit in (45) by a factor of 10 or so—still very far from the needed period of 35^d!

We conclude that our present knowledge of vortex pinning and of neutron stellar structure leaves us with only two distinct possibilities regarding the neutron star in Her X-1: Either its vortices are not pinned, or the 35^d period is not due to neutron stellar wobble.

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³ The following estimates do not differ significantly for different mass neutron stars.

REFERENCES

- Anderson, P. W. 1975, University of Cambridge preprint TCM/1975/5.
- Anderson, P. W., and Itoh, N. 1975, *Nature*, **256**, 25.
- Anderson, P. W., Pines, D., and Shaham, J. 1977, in preparation.
- Arnett, W., and Bowers, R. 1974, University of Texas Publications in Astronomy, No. 9.
- Bäckman, S., and Weise, W. 1975, *Phys. Letters*, **55B**, 1.
- Baym, G., Lamb, D. Q., and Lamb, F. K. 1975, University of Illinois preprint.
- Baym, G., Pethick, C. J., Pines, D., and Ruderman, M. 1969, *Nature*, **224**, 872.
- Brecher, K. 1972, *Nature*, **239**, 325.
- Giacconi, R. 1974, in *Proc. 16th Solvay Congress*, Université de Bruxelles, Bruxelles.
- Hoffberg, M., Glassgold, A., Richardson, R., and Ruderman, M. 1970, *Phys. Rev. Letters*, **24**, 775.
- Lamb, D. Q., Lamb, F. K., Pines, D., and Shaham, J. 1975, *Ap. J. (Letters)*, **198**, L21.
- Landau, L. D., and Lifshitz, E. M. 1959, *Fluid Mechanics* (London: Pergamon Press).
- Middleditch, J., Mast, T., and Nelson, J. 1975, 7th Texas Symposium on Relativistic Astrophysics, Dallas.
- Moiseyev, N. N., and Rumyantsev, V. V. 1968, *Dynamic Stability of Bodies Containing Fluid*, ed. N. H. Abramson (New York: Springer-Verlag).
- Negele, J., and Vautherin, D. 1973, *Nucl. Phys.*, **A207**, 298.
- Pandharipande, V. R., Pines, D., and Smith, R. A. 1976, University of Illinois, preprint.
- Pines, D., Pethick, C. J., and Lamb, F. K. 1973, *Ap. J.*, **184**, 271.
- Pines, D., and Shaham, J. 1972, *Phys. Earth Plan. Int.*, **6**, 103.
- Ruderman, M. A. 1976, *Ap. J.*, **203**, 213.
- Ruderman, M. A., and Sutherland, P. 1974, *Ap. J.*, **190**, 137.
- Shaham, J. 1975, preprint.
- . 1976a, Proc. HEAD meeting, January.
- . 1976b, *Phys. Letters*, **56A**, 461.

JACOB SHAHAM: Racah Institute of Physics, Hebrew University, Jerusalem, Israel